

Breast Cancer Risk Prediction Using Bayesian Logistic Regression

Sebastian Dustin Kumala¹, Annabelle Frederica Suryana², Chelsy Wandellice³, Felicia Tiffany⁴.

Data Science, Bina Nusantara University, Indonesia^{1,2,3,4}

sebastian.kumala@binus.ac.id¹, annabelle.suryana@binus.ac.id²,
chelsy.wandellice@binus.ac.id³ felicia.tiffany@binus.ac.id⁴,

Bina Nusantara University, Anggrek Campus, Kebon Jeruk Raya St., No. 27,
Kebon Jeruk, West Jakarta 11530, Indonesia

ABSTRACT

Breast cancer remains one of the leading causes of cancer-related mortality among women worldwide, making early detection and recurrence risk prediction essential for effective treatment planning. This study applies Bayesian Logistic Regression to analyze breast cancer outcomes using clinical and pathological features from the Breast Cancer Dataset available on Kaggle. The dataset consists of patient-level attributes describing tumor characteristics and diagnostic information, with a binary outcome variable indicating cancer status. Unlike classical logistic regression, the Bayesian framework explicitly models parameter uncertainty through posterior distributions and allows the incorporation of prior information, which is particularly advantageous for medical datasets with limited sample sizes and categorical predictors. A Bayesian logistic regression model was specified using weakly informative normal priors and estimated via Markov Chain Monte Carlo (MCMC) sampling. Model convergence was assessed using Gelman-Rubin and Geweke diagnostics, both of which indicate satisfactory convergence and stable posterior estimates. Posterior inference shows that several tumor-related features have a meaningful influence on breast cancer outcomes, as indicated by credible intervals that exclude zero. Model performance was evaluated using posterior predictive checks and classification metrics. The model achieved high predictive performance, with an accuracy of 94.55% and an area under the ROC curve (AUC) of 0.9867, demonstrating strong discriminative capability. Overall, the results suggest that Bayesian Logistic Regression provides an effective and reliable way for breast cancer detection to support clinical diagnosis.

Keywords: Bayesian Logistic Regression, Breast Cancer, MCMC, Posterior Inference

INTRODUCTION

Breast cancer is one of the most prevalent malignancies affecting women globally and continues to pose a significant public health challenge. Despite advances in medical technology and treatment strategies, accurate prediction of breast cancer outcomes remains crucial for early intervention, personalized treatment, and improved patient survival. As a result, data-driven approaches have become increasingly important in supporting clinical decision-making.

Recent developments in data science have enabled the application of statistical and machine learning techniques to medical datasets containing patient clinical and pathological information. Among these methods, logistic regression has been widely used for binary classification tasks due to its simplicity and interpretability. However, classical logistic regression relies on point estimates and large-sample assumptions, which may limit its effectiveness in medical datasets characterized by uncertainty, categorical variables, and relatively small sample sizes.

Bayesian methods provide an alternative modeling framework that addresses these limitations by incorporating prior knowledge and producing probabilistic interpretations of model parameters. Bayesian Logistic Regression, in particular, is well-suited for binary outcome variables and allows uncertainty to be explicitly modeled through posterior distributions. This makes it especially appropriate for medical prediction tasks where interpretability and uncertainty quantification are essential.

In this study, a breast cancer dataset obtained from Kaggle is analyzed using Bayesian Logistic Regression. The dataset contains clinical and tumor-related attributes that are commonly used in breast cancer diagnosis and prognosis. The objective of this research is to evaluate how these attributes influence breast cancer outcomes while leveraging the Bayesian framework to provide reliable predictions and meaningful uncertainty estimates.

LITERATURE REVIEW

Breast cancer prediction and classification have been extensively studied using a variety of statistical and machine learning techniques. Early research primarily relied on classical statistical models such as logistic regression and discriminant analysis, which offered interpretable results and were well-suited for structured clinical data. These methods demonstrated reasonable

performance in predicting breast cancer outcomes but often lacked robust mechanisms for handling parameter uncertainty and variability in small datasets.

With the advancement of machine learning, more complex models such as decision trees, support vector machines, random forests, and artificial neural networks have been widely applied to breast cancer datasets. While these approaches often achieve high predictive accuracy, they are frequently criticized for their limited interpretability, sensitivity to hyperparameter tuning, and reliance on large training datasets. These limitations pose challenges in medical research contexts, where transparency, explainability, and limited sample sizes are common concerns.

Bayesian approaches have gained increasing attention in healthcare analytics due to their ability to explicitly model uncertainty and incorporate prior medical knowledge into the learning process. Unlike frequentist methods that produce point estimates, Bayesian methods generate posterior distributions for model parameters, allowing uncertainty to be quantified through credible intervals. This characteristic is particularly valuable in clinical decision-making, where understanding confidence levels associated with predictions is as important as predictive accuracy itself.

Previous studies have demonstrated that Bayesian Logistic Regression performs comparably to, and in some cases better than, classical logistic regression for medical classification tasks, especially when dealing with categorical predictors and small-to-moderate sample sizes. The Bayesian framework also enhances interpretability by providing probabilistic interpretations of predictor effects, which supports more informed clinical insights.

To efficiently estimate posterior distributions in Bayesian models, modern sampling algorithms such as Hamiltonian Monte Carlo (HMC) have been widely adopted. HMC improves upon traditional Markov Chain Monte Carlo (MCMC) methods by utilizing gradient information of the posterior distribution, enabling faster convergence and more efficient exploration of high-dimensional parameter spaces. However, HMC requires careful tuning of hyperparameters such as step size and trajectory length.

The No-U-Turn Sampler (NUTS) is an extension of HMC that automatically adapts these hyperparameters during sampling, eliminating the need for manual tuning. By dynamically

determining the optimal trajectory length and preventing redundant sampling paths, NUTS improves sampling efficiency and stability. As a result, HMC with NUTS has become a standard inference method in Bayesian statistical modeling, particularly in probabilistic programming frameworks such as Stan and PyMC.

In the context of breast cancer analysis, Bayesian methods have been applied to diagnosis, recurrence prediction, and survival analysis. These studies consistently highlight the strengths of Bayesian inference in modeling uncertainty, improving interpretability, and supporting robust clinical decision-making. Based on the existing literature, the use of Bayesian Logistic Regression with HMC and No-U-Turn sampling is a well-justified and effective approach for analyzing the Kaggle breast cancer dataset and modeling breast cancer outcomes.

METHODOLOGY

3.1 Dataset Description

3.1.1 Dataset Origin

The dataset used in this study is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, originally obtained from the University of Wisconsin–Madison and publicly available through the UC Irvine Machine Learning Repository. A widely used version of this dataset is also available on Kaggle. The dataset contains features computed from digitized images of fine needle aspirates (FNA) of breast masses, with the objective of classifying tumors as malignant or benign.

3.1.2 Analysis Methodology

Since the target variable is binary, indicating whether a patient has breast cancer (malignant) or not (benign), the dataset will be analyzed using classification techniques. In this context, Bayesian Logistic Regression can be applied to estimate the probability of malignancy based on the provided features.

3.1.3 Variables in the Dataset Explanation

The dataset contains 32 columns, where 10 of the columns include an identifier and nine attributes describing tumor and patient characteristics. Each variable type and role is summarized below:

Variable	Role	Type	Description	Null Values
id	Feature	Integer	Patient ID	no
diagnosis	Target	Categorical	M = malignant, B = benign	no
radius_mean	Feature	Numeric	Mean of distances from center to points on the tumor boundary	no
texture_mean	Feature	Numeric	Standard deviation of gray-scale values	no
perimeter_mean	Feature	Numeric	Mean perimeter of the tumor	no
area_mean	Feature	Numeric	Mean area of the tumor	no
smoothness_mean	Feature	Numeric	Mean local variation in radius lengths	no
compactness_mean	Feature	Numeric	Mean compactness ($\text{perimeter}^2 / \text{area} - 1.0$)	no
concavity_mean	Feature	Numeric	Mean severity of concave portions of the contour	no
concave points_mean	Feature	Numeric	Mean number of concave portions of the contour	no
symmetry_mean	Feature	Numeric	Mean of symmetry values of the tumor cell nuclei	no
fractal_dimension_mean	Feature	Numeric	Mean fractal dimension of the tumor cell nuclei boundary	no

In addition to the mean measurements, the dataset includes corresponding standard error (_se) and worst (_worst) values for each tumor characteristic (e.g., radius_se,

area_worst), totaling 30 numeric features for each instance. The standard error feature captures the variability of the tumor property across multiple sampled regions within the same tumor image. These variables provide information about intra-tumor heterogeneity, which is clinically relevant because malignant tumors tend to exhibit greater structural irregularity and variability compared to benign tumors.

The worst features correspond to the largest (most extreme) observed values of each measurement among the sampled regions. These variables are designed to capture the most aggressive or abnormal portions of the tumor. From a clinical perspective, malignant tumors often present localized regions with extreme shape, size, or texture characteristics, making the worst-case measurements highly informative for distinguishing malignant from benign cases.

Together, the mean, standard error, and worst feature groups provide a multi-scale representation of tumor morphology, summarizing central tendency, variability, and extremal behavior. This comprehensive characterization enables classification models, including Bayesian Logistic Regression, to better capture subtle structural differences associated with breast cancer malignancy and improves both predictive performance and interpretability.

3.1.4 Dataset Source

Zwitter, M. & Soklic, M. (1988). Breast Cancer Dataset. UC Irvine Machine Learning Repository. Available on Kaggle: [Breast Cancer Dataset](#)

3.2 Model specification (Bayesian Logistic Regression)

Bayesian Logistic Regression (BLR) is a statistical modeling approach, an extension of classical logistic regression, that applies Bayesian inference to estimate model parameters. It is used when the response variable is binary, typically coded as 0 (failure) and 1 (success). The objective of BLR is to model the relationship between one or more predictor variables and the probability of success, while incorporating prior information about the parameters. In this study, the response variable represents tumor malignancy status (1 = malignant, 0 = benign), while the explanatory

variables consist of standardized tumor characteristics, namely radius mean, texture mean, and perimeter mean.

Unlike classical logistic regression, which estimates parameters using Maximum Likelihood Estimation (MLE), Bayesian Logistic Regression adopts a Bayesian framework, where uncertainty about model parameters is explicitly modeled using probability distributions. In this framework, unknown parameters are treated as random variables rather than fixed quantities. This makes BLR more flexible, particularly in cases of small sample sizes, separation issues, or when incorporating prior domain knowledge. The foundation of Bayesian inference is Bayes' theorem, which combines prior knowledge with information from the observed data. The Bayesian framework consists of three core components:

$$p(\beta | y) \propto p(y | \beta) p(\beta)$$

where:

- $p(\beta | y)$ is the posterior distribution
- $p(y | \beta)$ is the likelihood
- $p(\beta)$ is the prior distribution

The goal of Bayesian Logistic Regression is to estimate the posterior distribution of the regression coefficients, which represents all available information about the parameters after observing the data. In this model, regression coefficients are interpreted on the log-odds scale, such that a positive coefficient indicates an increase in the odds of malignancy as the corresponding predictor increases.

3.2.1 Prior Distribution (Normal Weakly Informative Prior)

The prior distribution represents information about the regression parameters before observing the data. In Bayesian logistic regression, regression coefficients β are commonly assigned Normal priors. A weakly informative prior is often chosen when there is limited prior knowledge but extreme parameter values are considered implausible. A typical specification is:

$$\beta_i \sim N(0, \sigma^2)$$

where the mean is set to zero and the variance is large (e.g., $\sigma^2 = 1000$ or larger). This choice allows the data to dominate the inference while providing mild regularization that helps stabilize estimation.

3.2.2 Likelihood (Bernoulli with Logit Link)

Since the response variable is binary, the likelihood is modeled using a Bernoulli distribution.

$$y_i \sim \text{Bernoulli}(\pi_i)$$

where π_i is the probability of success for observation i

The probability π_i linked to the predictors using the logit link function:

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip}$$

This transformation maps probabilities in the interval (0, 1) to the entire real line, allowing a linear relationship between predictors and the log-odds of the outcome.

Assuming independence across observations, the full likelihood for n observations is:

$$\prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i}$$

3.2.3 Posterior Distribution

The posterior distribution is obtained by multiplying the prior distribution with the likelihood:

$$p(\beta | y) \propto p(y | \beta) p(\beta) = p(\beta | y) \propto \prod_{i=1}^n \pi_i^{y_i} (1 - \pi_i)^{1-y_i} \times p(\beta)$$

The posterior contains all updated knowledge about the model parameters after observing the data. In Bayesian logistic regression, this posterior does not have a closed-form solution, which necessitates the use of simulation-based methods for inference.

3.3 Model Inference (MCMC)

Since the posterior distribution in Bayesian logistic regression is analytically intractable, Markov Chain Monte Carlo (MCMC) methods are used to approximate it. MCMC generates samples from the posterior distribution by constructing a Markov chain whose stationary distribution is

the target posterior. These samples allow empirical estimation of posterior summaries such as means, standard deviations, and credible intervals. Common MCMC algorithms for Bayesian logistic regression include Hamiltonian Monte Carlo (HMC), a gradient-based MCMC method that efficiently explores high-dimensional posterior distributions, with No-U-Turn Sampler (NUTS), an extension of HMC that automatically tunes step sizes and trajectory lengths. HMC-based methods are particularly efficient for logistic regression models because they use gradient information to explore the posterior more effectively and avoid random-walk behavior.

Parameter estimates are typically obtained using the posterior mean or median of the MCMC samples. Statistical significance is assessed using credible intervals. A regression coefficient is considered to have a meaningful effect if the posterior credible range (25%-97.5% posterior quantiles) does not include zero. This Bayesian approach provides a probabilistic interpretation of uncertainty and offers greater flexibility compared to classical logistic regression.

RESULTS & DISCUSSION

This section presents the results of the Bayesian logistic regression analysis and a detailed discussion of the model estimation, posterior behavior, and predictive performance.

4.1 Model Estimation Results

Parameter estimation for the Bayesian logistic regression model was conducted using Markov Chain Monte Carlo (MCMC) sampling. Multiple chains were generated and combined to obtain posterior samples for the intercept (α) and regression coefficients (β_1 - β_{10}), corresponding to the selected tumor-related predictors. All predictors were standardized prior to model estimation to ensure comparability of effect sizes and stable posterior sampling.

The convergence and sampling behavior of the MCMC chains were first examined visually using trace plots and posterior density plots. Figure 4.1.a displays the trace plots (left panels) and posterior density plots (right panels) for the intercept and all regression coefficients. The trace plots exhibit stable oscillations around constant mean levels across iterations, with no evident trends or long term drifts. This pattern indicates good mixing and suggests that the Markov chains have reached stationarity. The corresponding posterior density plots are smooth and unimodal, implying that the posterior distributions are well identified and that parameter estimates are stable.

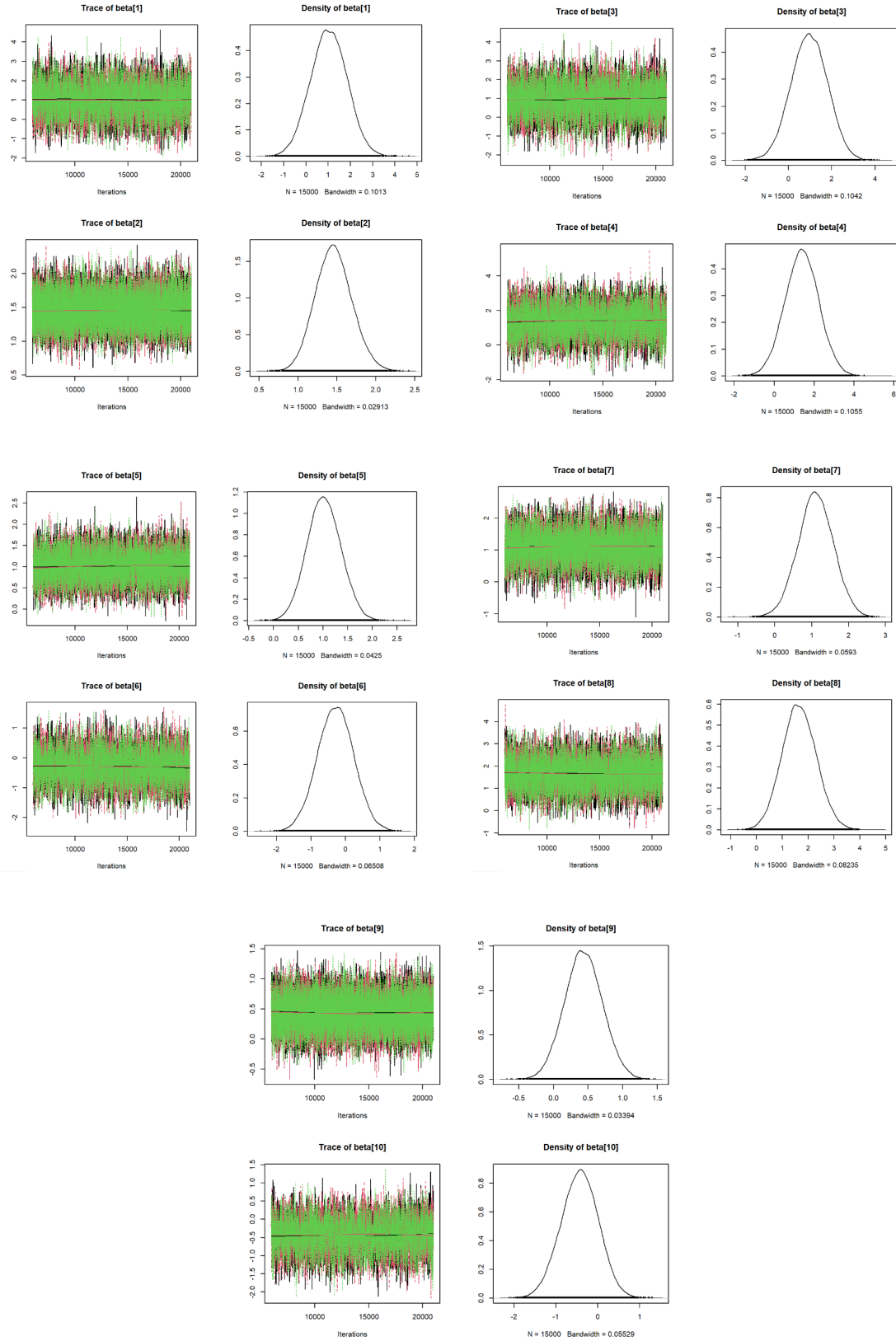


Fig. 4.1.a Trace plots and posterior density plots for the intercept and regression coefficients.

Following the visual assessment, numerical summaries of the posterior distributions were computed to quantify parameter uncertainty. Posterior quantiles (25%, 75%, and 97.5%) were used to summarize the central tendency and dispersion of each parameter. These summaries are reported in Figure 4.1.b, which presents the posterior result table for the intercept and regression coefficients.

The posterior summary indicates that the intercept (α) is concentrated on negative values, reflecting a low baseline log-odds of malignancy when predictors are held at their standardized reference levels. Among the regression coefficients (β_1 - β_{10}) several predictors, such as radius mean, texture mean, perimeter mean, area mean, smoothness mean, concavity mean, concave points mean, and symmetry mean, exhibit posterior distributions that are predominantly positive, suggesting positive associations with the probability of malignancy. In contrast, compactness mean (β_6) and fractal dimension mean (β_{10}) show posterior distributions that span both negative and positive values, indicating greater uncertainty regarding the direction and strength of their effects.

	25%	75%	97.5%
alpha	-0.7397173	-0.44307972	-0.1567619
beta[1]	0.4760613	1.58105402	2.6186403
beta[2]	1.2986154	1.61250511	1.9366818
beta[3]	0.3951012	1.53170099	2.5734221
beta[4]	0.8128375	1.95407490	3.0501690
beta[5]	0.7879905	1.24878317	1.6979727
beta[6]	-0.6427112	0.06856243	0.7400516
beta[7]	0.7913220	1.43029290	2.0461699
beta[8]	1.2074804	2.10352915	2.9826026
beta[9]	0.2497773	0.61780713	0.9739884
beta[10]	-0.7310549	-0.12865603	0.4191266

Fig 4.1.b. Posterior Result Summary

Overall, the posterior estimation results suggest that the Bayesian logistic regression model successfully captures meaningful relationships between the explanatory variables and the binary

malignancy outcome, while also appropriately reflecting uncertainty for predictors with weaker or less stable effects. These results provide a coherent foundation for the detailed posterior interpretation and credible interval analysis presented in the following section.

4.2 Posterior Distribution and Credible Interval Analysis

This section presents a Bayesian interpretation of the posterior distributions of the logistic regression parameters. The analysis focuses on the direction, magnitude, and posterior uncertainty of each effect, as inferred from posterior quantile summaries. In particular, inference is based on the central posterior mass captured by the 25%-75% posterior quantiles, with the 97.5% posterior quantile reported to characterize upper-tail uncertainty. This representation highlights the stability of the core posterior estimates while retaining information about the potential extent of larger effect sizes. Rather than relying on frequentist confidence intervals, all conclusions are drawn from the concentration and shape of the posterior probability distributions.

Intercept (α)

The posterior distribution of the intercept is concentrated on negative values. The central posterior mass, defined by the 25%-75% posterior quantiles, spans approximately from -0.74 to -0.44, with the upper posterior tail extending to -0.16 (97.5% posterior quantile). This indicates that, when all predictors are held at their standardized reference levels, the baseline log-odds of malignancy are negative. From a Bayesian perspective, this implies a low baseline probability of malignancy in the absence of strong tumor-related features. The unimodal and tightly concentrated posterior distribution suggests a stable and well-identified intercept parameter.

Radius Mean (β_1)

The posterior distribution of the coefficient for radius mean is predominantly positive. The central posterior mass (25%-75% quantiles) lies approximately between 0.48 and 1.58, with a long upper tail reaching about 2.62. The absence of substantial posterior mass near zero provides strong Bayesian evidence that larger tumor radius increases the probability of malignancy. The relatively wider spread of the posterior indicates moderate uncertainty in the magnitude of the effect, while its direction remains well supported.

Texture Mean (β_2)

The coefficient for texture mean exhibits a strongly positive posterior effect with low uncertainty. The central posterior mass spans approximately from 1.30 to 1.61, and the posterior tail extends to around 1.94. The sharply peaked posterior density indicates high parameter identifiability and strong Bayesian evidence that texture-related tumor characteristics play a critical role in distinguishing malignant from benign cases.

Perimeter Mean (β_3)

The posterior distribution for perimeter mean is strictly positive, with the central posterior mass ranging approximately from 0.40 to 1.53 and the upper posterior tail reaching about 2.57. This concentration of posterior probability away from zero indicates a substantial and credible increase in the log-odds of malignancy as tumor perimeter increases, confirming perimeter mean as an important structural predictor in the Bayesian model.

Area Mean (β_4)

The coefficient associated with area mean shows a strong positive posterior effect. The central posterior mass (25%-75% quantiles) ranges approximately from 0.81 to 1.95, with a pronounced upper tail extending to about 3.05. This suggests that larger tumor area is strongly associated with higher malignancy risk, and that tumor size is a dominant factor in the posterior predictive structure of the model.

Smoothness Mean (β_5)

The posterior distribution for smoothness mean is entirely positive. The central posterior mass spans approximately from 0.79 to 1.25, with the posterior tail extending to around 1.70. This indicates consistent Bayesian support for a positive association between surface smoothness and malignancy, although the magnitude of the effect is moderate relative to size and concavity related predictors.

Compactness Mean (β_6)

The posterior distribution of compactness exhibits substantial uncertainty, with posterior mass distributed across both negative and positive values. The central posterior mass is concentrated near zero, with the 75% posterior quantile only slightly above zero, and posterior tails extending from approximately -0.64 to 0.74. This pattern indicates weak Bayesian evidence for an

independent effect of compactness on malignancy once other tumor characteristics are accounted for in the model.

Concavity Mean (β_7)

The coefficient for concavity mean shows a clearly positive posterior distribution. The central posterior mass ranges approximately from 0.79 to 1.43, with the upper posterior tail reaching about 2.05. This indicates strong Bayesian support for an increased probability of malignancy as tumor concavity increases, highlighting concavity as a meaningful morphological marker.

Concave Points Mean (β_8)

The posterior distribution of concave points displays one of the strongest positive effects in the model. The central posterior mass lies approximately between 1.21 and 2.10, with the posterior tail extending to about 2.98. The high concentration of posterior probability away from zero and the relatively narrow uncertainty indicate that concave points are a highly informative predictor of malignancy in the Bayesian framework.

Symmetry Mean (β_9)

The posterior distribution for symmetry mean is positive, with the central posterior mass spanning approximately from 0.25 to 0.62 and an upper tail reaching about 0.97. Although the magnitude of the effect is smaller compared to size-related predictors, the posterior distribution consistently favors a positive association with malignancy.

Fractal Dimension Mean (β_{10})

The posterior distribution for fractal dimension mean shows substantial uncertainty, with posterior mass distributed around zero and tails extending from approximately -0.73 to 0.42. This indicates insufficient Bayesian evidence to support an independent association between fractal dimension and malignancy in the presence of other tumor characteristics.

4.3 Predictive Performance Evaluation

The predictive performance of the Bayesian logistic regression model was evaluated using posterior predictive checks and classification-based performance metrics. This evaluation aims to assess both the adequacy of the model in reproducing the observed data and its ability to discriminate between malignant and benign cases based on posterior predictive probabilities.

4.3.1 Posterior Predictive Checks

Posterior predictive checks (PPC) were conducted to examine whether the Bayesian logistic regression model is capable of generating data that are consistent with the observed outcomes. Using the posterior distribution of the predicted probabilities, replicated binary outcomes were generated and compared with the observed diagnostic labels.

The results indicate that the distribution of predicted malignancy proportions closely matches the observed proportion in the dataset. This agreement suggests that the model is well-calibrated at the population level and does not systematically overestimate or underestimate the prevalence of malignant cases. The close alignment between observed and replicated summaries provides evidence that the model adequately captures the underlying data generating process and supports the validity of subsequent predictive evaluations.

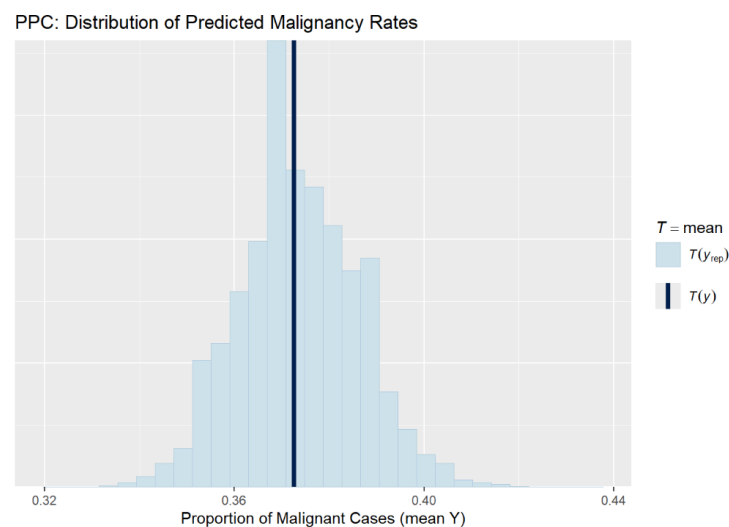


Fig. 4.3.1. Posterior predictive checks

4.3.2 Classification Performance

The discriminative performance of the model was evaluated using classification based metrics derived from the posterior mean of the predicted probabilities. A probability threshold of 0.5 was applied to convert probabilistic predictions into binary class labels.

A. Confusion Matrix and Accuracy

The confusion matrix summarizes the classification outcomes across predicted and actual diagnostic categories. Using the posterior mean predictions, the model correctly classified

348 benign cases and 190 malignant cases. A total of 9 benign cases were incorrectly classified as malignant (false positives), while 22 malignant cases were incorrectly classified as benign (false negatives). These results indicate that the model achieves strong overall performance, with a relatively small number of misclassifications. In particular, the reduced number of false negatives suggests that the model is effective in identifying malignant cases, which is critical in a clinical screening context.

	Predicted	
Actual	0	1
0	348	9
1	22	190

Fig. 4.3.2.a. Confusion Matrix Across Predicted and Actual Classes

The overall classification accuracy achieved by the model is 94.55%, indicating that the majority of observations are correctly classified. This high accuracy demonstrates that the posterior mean probabilities provide a reliable basis for class assignment. However, accuracy alone does not fully capture discriminative performance, especially under different classification thresholds.

To provide a more clinically interpretable evaluation, sensitivity and specificity were also considered. The model achieved a sensitivity of 89.62%, reflecting its ability to correctly identify malignant cases, and a specificity of 97.48%, indicating strong performance in correctly classifying benign cases. These results suggest that the model maintains a favorable balance between detecting malignancies and minimizing false alarms.

B. Receiver Operating Characteristic (ROC) Curve and AUC

To further assess discriminative performance independently of a fixed classification threshold, the receiver operating characteristic (ROC) curve was constructed by varying the probability cutoff across its full range. The area under the ROC curve (AUC) was computed as a summary measure of separability between malignant and benign cases. The model achieved an AUC value of 0.9867, indicating excellent discriminative ability. This near perfect AUC demonstrates that the posterior predictive probabilities are highly

effective in ranking observations according to their likelihood of malignancy across a wide range of decision thresholds.

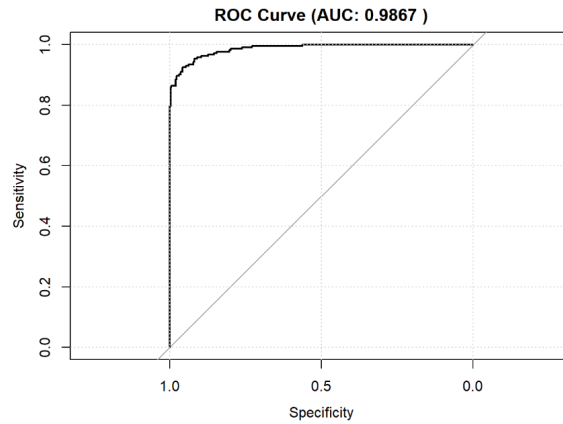


Fig. 4.3.2.b. ROC Curve Across Probability Thresholds

4.4 MCMC Convergence Diagnostics

4.4.1 Gelman-Rubin Diagnostics

Convergence of the MCMC simulations was evaluated using the Gelman-Rubin potential scale reduction factor (PSRF). This diagnostic checks convergence by comparing the variability of parameter estimates within individual chains compared to multiple chains with different initial values. As the number of samples increases and chains converge to the same posterior distribution, the between-chain and within-chain variances become similar. PSRF close to 1 means convergence, meanwhile values far greater than 1 mean more sampling is required.

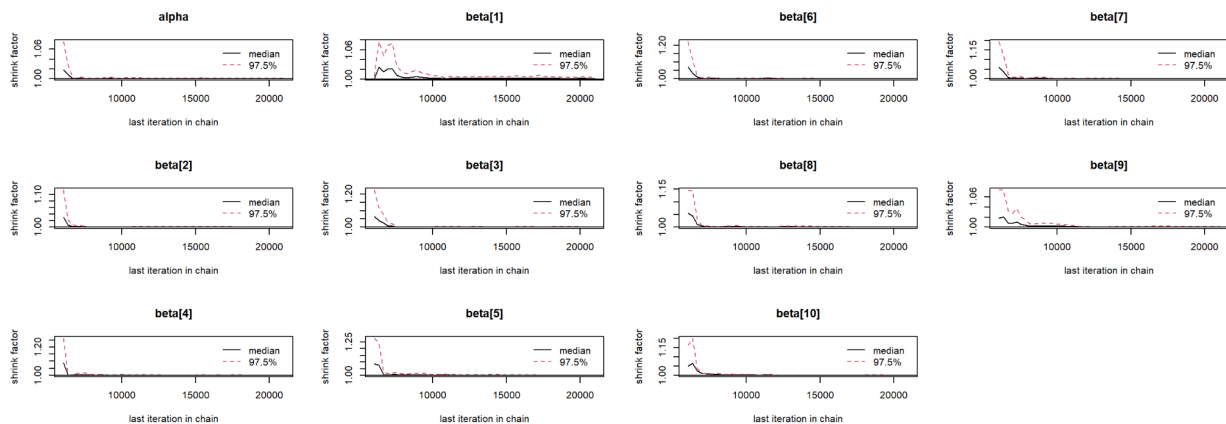


Fig. 4.4.1.a. Gelman-Rubin PSRF Plots

Figure 4.4.1.a. shows the change of the PSRF for α and β_1 until β_{10} along all iterations. The line represents the median PSRF across chains, while the dashed line represents the 97.5% upper confidence limit for a more conservative measure of convergence. In early stages of sampling, all parameters exhibit PSRF values exceeding 1, meaning incomplete convergence and the significant influence from initial values. However, as the number of iterations increase, the PSRF values decrease sharply then stabilize near 1. By around 8,000 - 10,000 iterations, both the median and upper confidence limits of the PSRF for all parameters are equal to 1, which means convergence occurred across all chains.

```
## Potential scale reduction factors:
##
##      Point est. Upper C.I.
## alpha          1      1
## beta[1]         1      1
## beta[2]         1      1
## beta[3]         1      1
## beta[4]         1      1
## beta[5]         1      1
## beta[6]         1      1
## beta[7]         1      1
## beta[8]         1      1
## beta[9]         1      1
## beta[10]        1      1
##
## Multivariate psrf
##
## 1
```

Fig. 4.4.1.b. Gelman-Rubin PSRF Summary

Figure 4.4.1.b. shows the PSRF summary also further supports the same conclusion. The point estimates and upper CI for all parameters, also the multivariate PSRF all equal to 1. This means convergence has happened in all parameters. These two results provide strong evidence that the MCMC chains have converged to a common posterior distribution.

4.4.2 Geweke Diagnostics

Convergence was further assessed using Geweke's diagnostic, which evaluates within-chain stationarity by comparing the mean of an early segment of the Markov Chain Monte Carlo (MCMC) sample (first 10%) with the mean of a later segment (last 50%). The diagnostic computes standardized Z-scores for each parameter; values within ± 1.96 indicate no statistically significant difference between the early and late portions of the chain, providing evidence that the chain has reached its stationary distribution.

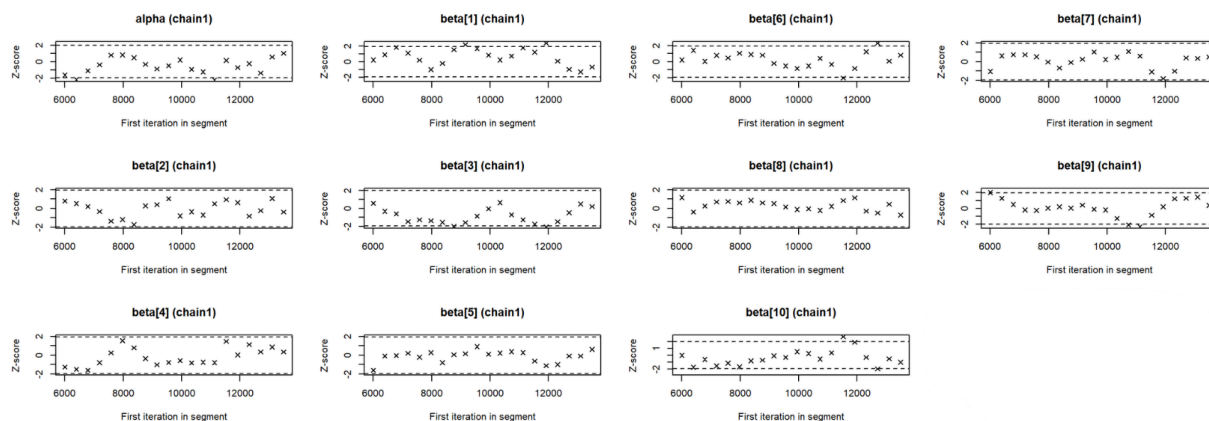


Fig. 4.4.2.a. Geweke Plots of Chain 1

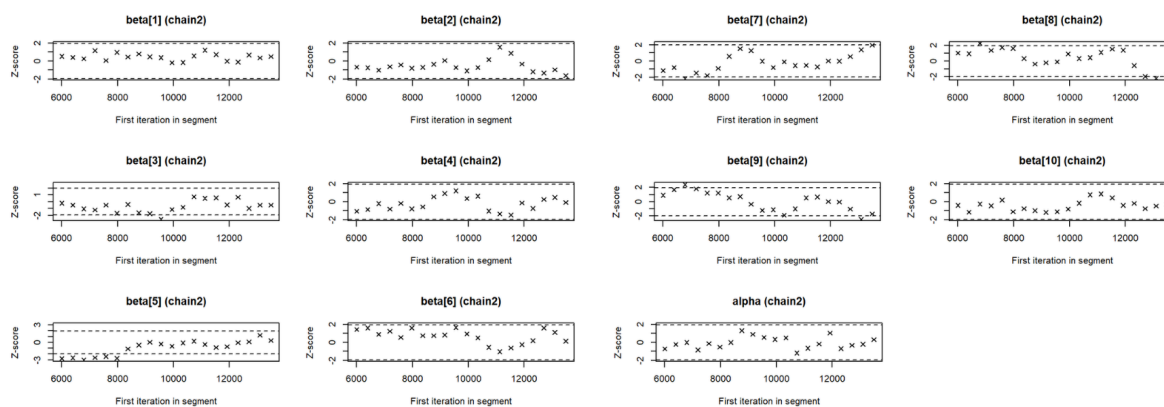


Fig. 4.4.2.b. Geweke Plots of Chain 2

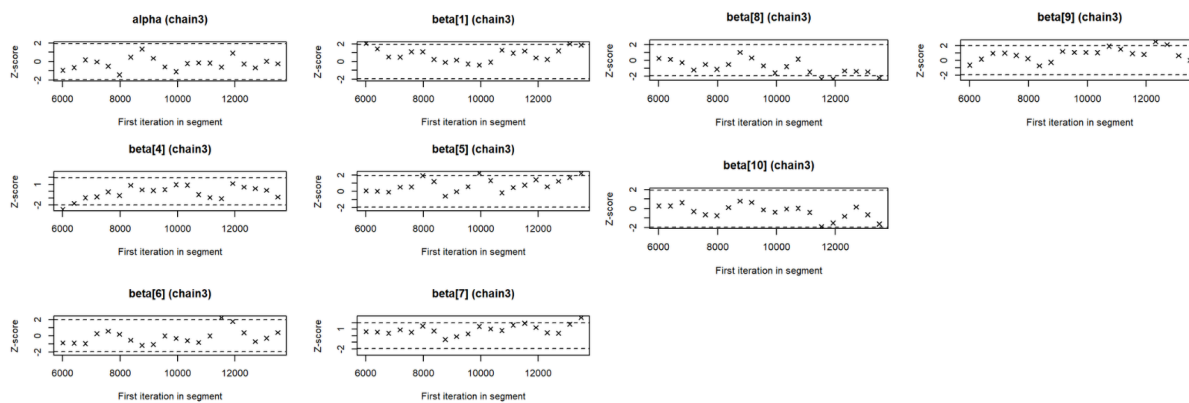


Fig. 4.4.2.c. Geweke Plots of Chain 3

Figure 4.4.2.a. until 4.4.2.c. shows the Geweke Plots of three chains with different initial values across the iterations of the model. In each parameter and chain, the Z-scores fluctuate randomly around zero, but remain within the confidence bounds. This means there are no strong trends or sustained deviations over a long period of the iterations. While there are a couple of above the threshold in some iterations, the amount only shows fluctuations of the model. There is no significant structure change or directional drift in these plots. Meaning, the chains have stabilized and are sampling from a stationary posterior distribution..

```
## [[1]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##   alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## -1.62128  0.22786  0.79675  0.56690 -1.26296 -1.59959  0.20674 -1.03038
##   beta[8]  beta[9]  beta[10]
##   1.16065  2.00756 -0.02343
##
##
## [[2]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##   alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
##  -0.7048  0.5277  -0.6547  -0.2161  -1.0601  -2.7617  1.4237  -1.1738
##   beta[8]  beta[9]  beta[10]
##   1.0189  0.9129  -0.4013
##
##
## [[3]]
##
## Fraction in 1st window = 0.1
## Fraction in 2nd window = 0.5
##
##   alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## -0.92512  2.05746 -0.40045  0.42410 -2.62820  0.07981 -0.86303  0.65103
##   beta[8]  beta[9]  beta[10]
##   0.24359 -0.62869  0.27771
```

Fig. 4.4.2.d. Geweke Summary across 3 Chains

Figure 4.4.2.d. shows the Geweke summary for all parameters in each chain. Across all three chains, the Geweke Z-scores for the parameters α and β_1 until β_{10} well within the range of the ± 1.96 threshold. This indicates there are no statistically significant differences between the early and late segments of the chains. No parameters that exhibit extreme values suggest non-convergence of the model. The similarity of Geweke diagnostics across the chains also supports the proof of convergence. Since there are no significant outliers or sign reversals in Z-scores across chains, the posterior distributions can be seen as chain independent.

4.4.3 Effective Sample Size (ESS)

The Effective Sample Size (ESS) calculates the number of independent samples contained in the MCMC. Higher ESS values indicate better chain mixing and more reliable posterior estimates.

```
##      alpha  beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]
## 13395.127 4400.467 14243.253 4467.096 4753.361 6796.169 4404.541 6133.744
##  beta[8]  beta[9]  beta[10]
## 6400.796 13667.607 3516.459
```

Fig. 4.4.3.a. ESS Summary

Figure 4.4.3.a shows the ESS summary of all parameters from the model. All parameters show an ESS of above 1000, which is often used as the baseline for stable estimates. Meaning that the model produced a good amount of independent samples and a relatively low autocorrelation. Considering the number of iterations used for the model is 15000, the ESS values of all parameters indicate that around ~25-35% of the samples are independent. Most parameters retain a large proportion of independent information from the MCM sample. Overall, the MCMC sampling can be considered efficient and converged, and the posterior summaries for all parameters are statistically reliable.

CONCLUSION

This study applied Bayesian Logistic Regression (BLR) to predict breast cancer malignancy using clinical and tumor-related features from the Ljubljana Breast Cancer dataset. The Bayesian framework enabled explicit uncertainty quantification through posterior distributions estimated using Markov Chain Monte Carlo (MCMC) sampling, with convergence confirmed using multiple diagnostic measures, including trace plots, Gelman-Rubin (PSRF) and Geweke diagnostics, show that all parameters converged to a common posterior distribution. These results confirm that the posterior samples are reliable and suitable for inference.

Through posterior analysis, it was revealed that all tumor-related predictors included in the model resulted in positive and statistically meaningful effects on the probability of malignancy. For each coefficient, the 95% credible interval excluded zero, providing strong Bayesian evidence that these features contribute significantly to malignancy risk. Among the predictors, texture mean demonstrated the most precise and stable posterior estimate, indicating that tumor texture characteristics play a particularly important role in distinguishing malignant from benign

cases. The intercept was consistently negative, implying a relatively low baseline probability of malignancy when tumor characteristics are held at their standardized reference levels.

Model adequacy was further supported through posterior predictive checks, which showed close agreement between observed and replicated malignancy proportions. This indicates that the Bayesian logistic regression model is well calibrated and capable of reproducing key characteristics of the observed data, supporting its validity as a generative model for breast cancer outcomes.

In terms of predictive performance, the Bayesian model demonstrated strong discriminative ability. Using posterior mean predicted probabilities, the model achieved high classification accuracy and a favorable balance between sensitivity and specificity. The confusion matrix showed a relatively low number of false negatives, which is particularly important in a clinical screening context where failing to identify malignant cases can have severe consequences. Additionally, the Receiver Operating Characteristic (ROC) analysis yielded an AUC close to 1, indicating excellent separability between malignant and benign cases across a wide range of decision thresholds.

Overall, the findings demonstrate that Bayesian Logistic Regression provides an effective, interpretable, and robust framework for breast cancer malignancy prediction. Compared to classical logistic regression, the Bayesian approach offers substantial advantages by delivering full posterior distributions, credible intervals, and principled uncertainty quantification, rather than relying solely on point estimates and asymptotic assumptions. These properties make the Bayesian framework particularly well suited for medical datasets characterized by limited sample sizes and correlated clinical predictors.

FUTURE WORK

Limitations This study has limitations that should be acknowledged. The analysis is based on a limited set of features and a single dataset, which may not fully capture the complex factors on breast cancer risk. Also, although convergence diagnostics and effective sample sizes indicate a good posterior estimation, the model used assumes linear effects on the log-odds scale. This may oversimplify nonlinear relationships among features.

Recommendations for future research Future research can extend this work by using more risk factors to improve predictive performance. Testing other Bayesian models may better capture complex relationships among features. Using more datasets could also allow the clinical use of Bayesian breast cancer risk prediction models.

REFERENCES

- Aryani, N. K. L., Sumarjaya, I. W., & Sari, K. (2023). Penerapan Metode Regresi Logistik Bayes dalam Menentukan Faktor-Faktor yang Mempengaruhi Kurangnya Minat Masyarakat Menggunakan QRIS. *Journal on Education*, 5(3), 10377-10386. <https://jonedu.org/index.php/joe/article/view/1936>
- Acquah, H. D.-G. (2013). Bayesian Logistic Regression Modelling via Markov Chain Monte Carlo Algorithm. *Journal of Social and Development Sciences*, 4(4), pp. 193-197. <https://doi.org/10.22610/jsds.v4i4.751>
- Fabreti, L. G., & Höhna, S. (2022). Convergence assessment for Bayesian phylogenetic analysis using MCMC simulation. *Methods in Ecology and Evolution*, 13, 77–90. <https://doi.org/10.1111/2041-210X.13727>
- Gabry, J., Simpson, D., Vehtari, A., Betancourt, M., & Gelman, A. (2019). Visualization in Bayesian workflow. *Journal of the Royal Statistical Society Series A (Statistics in Society)*, 182(2), 389-402. <https://doi.org/10.1111/rssa.12378>
- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). *CRC Press*. <https://sites.stat.columbia.edu/gelman/book/BDA3.pdf>
- Lukman, P. A., Abdullah, S., & Rachman, A. (2021). Bayesian logistic regression and its application for hypothyroid prediction in post-radiation nasopharyngeal cancer patients. *Journal of Physics: Conference Series*, 1725(1), Article 012010. <https://doi.org/10.1088/1742-6596/1725/1/012010>
- Robin, X., Turck, N., Hainard, A., Tiberti, N., Lisacek, F., Sanchez, J. C., & Müller, M. (2011). pROC: an open-source package for R and S+ to analyze and compare ROC curves. *BMC Bioinformatics*, 12, 77. <https://doi.org/10.1186/1471-2105-12-77>